The Transcriptional Shockwave Carries the Partner: A Virtual-Cell Foundation Model for Synthetic-Lethal Co-Dependency Prediction Beyond Screened Genes

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Abstract

Synthetic lethality (SL) is a powerful route to selective cancer therapy, but the space of candidate gene pairs is far too large to screen exhaustively, so computational prioritization is essential. Existing predictors largely perform link prediction over a curated SL interaction graph; their signal lives in graph topology, so a degree-only probe can win the easiest evaluation split while accuracy collapses for genes absent from the training graph — exactly the cold-start regime where new biology lies. Our insight is that a gene's perturbation transcriptome, the genome-wide "shockwave" that already predicts its own dependency, should also encode its synthetic-lethal partners. We first confirm this on a SynLethDB-derived K562 benchmark: with the evaluation harness held fixed, adding the observed perturbation transcriptome more than doubles partner-ranking quality (NDCG@10 $0.042 \rightarrow 0.094$) when one partner is anchored by a real Perturb-seq profile, though the lift vanishes when both genes are unseen. Because an observed response exists only for screened genes, data coverage becomes the central bottleneck: even broad K562 GWPS data leave many candidate genes and pairs without measured response profiles. We then propose generating the missing response: an end-to-end framework that drives a frozen virtual-cell foundation model (STATE) with a protein-language-model gene identity (ESM2) through a trainable adapter, so a perturbation response — and an SL prediction — can be produced for genes outside the backbone's fixed vocabulary (83.7% of our universe). We report this framework as a preliminary method: it runs end to end and scores out-of-vocabulary genes, but does not yet clear a strong dependency-only floor. Finally, a decomposition shows that most genome-wide co-dependency signal is pan-essentiality, isolating pair-specific SL as the hard residual that a generative virtual-cell model must learn. All results are scoped to K562 and to a benchmark-adapter label, not a validated SL assay.

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1 Introduction

Synthetic lethality (SL) — where disrupting either of two genes is tolerated but disrupting both kills the cell — is one of the few mechanisms that turns a tumor's own genetics into a selective drug target, as the clinical success of PARP inhibitors in BRCA-mutant cancers demonstrates [O'Neil et al., 2017, Huang et al., 2019a]. The bottleneck is no longer whether SL can be exploited but which pairs to pursue: the human genome admits on the order of 10^8 candidate gene pairs, far beyond what combinatorial CRISPR screens can cover, and SL is context-specific, so a pair that is lethal in one cellular background need not be in another. Computational prioritization is therefore a prerequisite for experimental follow-up, and this paper focuses on the specific setting of ranking a gene's candidate SL partners within a single, well-characterized cellular context (K562).

Most existing predictors meet this need by link prediction over a curated SL interaction graph, and they are strong when a test pair shares genes with the training graph [Feng et al., 2024, Wang et al., 2022b]. Their signal, however, lives in graph topology: because positive pairs are enriched among high-degree hub genes, a probe using gene degree alone can win the easiest evaluation split, while accuracy collapses once one or both genes are absent from the training graph. This exposes the real challenge — true cold-start prediction for genes with no curated interactions and, in the hardest regime, no prior measurement at all. It is precisely the regime where undiscovered biology lives, and the regime where graph-topology methods have no signal to stand on.

Our starting observation is that a different signal source sidesteps the graph entirely. When a gene is silenced, the cell's genome-wide transcriptional response — a perturbation "shockwave" — reflects that gene's functional wiring; the same response that predicts a gene's own dependency [Chiu et al., 2021, Haider et al., 2025] should also carry information about the partners whose loss it cannot tolerate. We confirm this directly: with the evaluation harness held fixed, adding the observed perturbation transcriptome more than doubles partner-ranking quality in the regime where one partner is anchored by a real Perturb-seq profile (Figure 1). This is the empirical license for our approach, but it also reveals its limit — an observed response exists only for genes that were actually screened, so it cannot reach the cold-start frontier.

The observed-response limit is a data-coverage problem rather than a modeling detail. Even genome-scale Perturb-seq libraries measure only a subset of the SL candidate universe, and a pairwise task amplifies this missingness because both genes would ideally have response profiles. Thus an observed-transcriptome model can test the anchored CV2 regime, where at least one partner is measured, but it cannot be a complete answer to both-gene-cold CV3. We quantify this coverage gap in the experiments and treat it as the central reason to move from measured to generated responses.

We therefore propose generating the response instead of measuring it. Our second contribution is an end-to-end framework that drives a frozen virtual-cell foundation model (STATE; Adduri et al., 2025) with a protein-language-model gene identity (ESM2; Lin et al., 2023) through a trainable adapter, so that a perturbation response — and hence an SL prediction — can be produced for any gene, including the 83.7% of our benchmark universe that lies outside the backbone's fixed vocabulary. The insight is that if the *observed* shockwave encodes a gene's partners, a faithfully *generated* shockwave

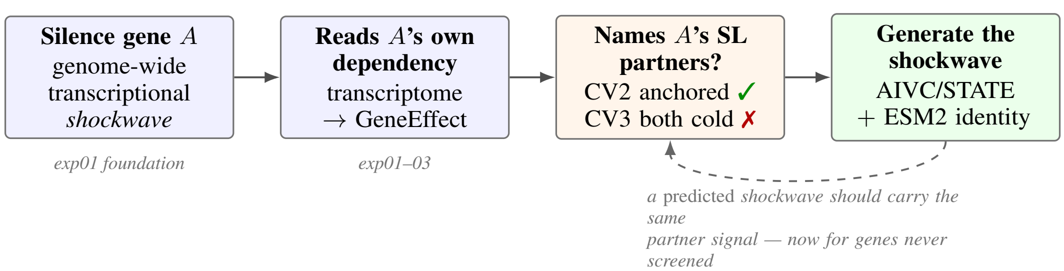

Figure 1: The transcriptional shockwave carries the partner. Perturbing a gene produces a genome-wide response that predicts the gene's own dependency (exp01). The same observed response also ranks a gene's synthetic-lethal partners when one partner is anchored by a real profile (CV2), but not when both genes are unscreened (CV3). Our contribution is a virtual-cell foundation model that *generates* this response from a protein-language-model gene identity, so the partner signal can in principle reach pairs neither gene was screened for (dashed loop; results preliminary).

should carry the same signal while extending it past the measured Perturb-seq library. We report this framework as a preliminary method: it runs end to end and produces predictions for out-of-vocabulary genes, but its current results do not yet clear a strong dependency-only baseline, and we are explicit about this throughout.

Our contributions are: (i) a leakage-controlled evaluation on a SynLethDB-derived K562 benchmark that separates an honest cold-start surface from a degree-gameable diagnostic, together with a dependency-only floor on it; (ii) the proof-of-concept that the observed perturbation transcriptome encodes a gene's SL partners when one partner is anchored, but is structurally limited by GWPS coverage; (iii) a virtual-cell framework that generates responses for unscreened genes via an ESM2-conditioned adapter to a frozen backbone, reported with preliminary results; and (iv) a decomposition showing that most genome-wide co-dependency signal is pan-essentiality, isolating pair-specific SL as the hard residual the generative approach must learn. All claims are scoped to K562 and to a benchmark-adapter label, not a validated SL assay.

2 Related Work

Computational synthetic-lethality prediction. Most SL predictors treat the problem as link prediction over a known SL interaction graph curated in databases such as SynLethDB [Wang et al., 2022a] and SLKB [Gokbag et al., 2023]. Matrix-factorization and graph-neural-network methods (GRSMF, SL2MF, DDGCN, and SLGNN; Huang et al., 2019b, Liu et al., 2020, Cai et al., 2020, Zhu et al., 2023), which we reproduce under our shared K562 protocol (Table 3), propagate observed SL labels through gene–gene similarity or interaction topology, and the unified benchmark of Feng et al. [2024] shows they perform strongly when test pairs share genes with the training graph. Their common weakness is the same as their strength: because the signal lives in graph topology and node degree, accuracy degrades sharply when one or both genes are unseen, and a degree-only probe can top the easiest split (our Table 2). We use that benchmark and its difficulty splits as our evaluation harness, but our features come from a gene's *perturbation response* rather than its position in the SL graph, which is what gives the approach a path to genes with no curated interactions.

Virtual-cell and perturbation-response models. A second line of work models the cell's response to perturbation directly. The virtual-cell program [Bunne et al., 2024] and the associated challenge [Roohani et al., 2025] frame this as learning to predict how a cell-state distribution shifts under genetic perturbation; STATE [Adduri et al., 2025] is a transformer that predicts such set-level responses, and single-cell foundation models such as scGPT [Cui et al., 2024] and Tahoe-x1 [Gandhi et al., 2025] pretrain transferable embeddings at scale. Two limitations matter for our setting. First, these models identify a perturbation through a fixed gene vocabulary, so out-of-vocabulary genes cannot be queried without an adapter — the bottleneck our ESM2-conditioned adapter [Lin et al., 2023] targets. Second, careful benchmarking finds that deep perturbation predictors do not yet consistently beat simple linear baselines [Ahlmann-Eltze et al., 2025, Vinas Torne et al., 2025], which is why we frame our generative framework as a preliminary method and compare it candidly against a strong dependency-only floor rather than claiming superiority.

Cancer dependency maps and the essentiality–SL distinction. A third line characterizes single-gene dependencies from genome-scale CRISPR screens. The DepMap effort, using CERES/Chronos correction [Meyers et al., 2017], quantifies how much each gene's loss impairs fitness, and downstream work builds clinically informed and translational dependency maps and target-prioritization frameworks [Pacini et al., 2024, Shi et al., 2024], including deep-learning prediction of dependency from tumor omics [Chiu et al., 2021]. We rely on these dependency scores as features and as a prediction target for a gene's own essentiality, but we are careful not to conflate essentiality with synthetic lethality: a broadly essential gene is not an SL partner, and our decomposition shows that genome-wide co-dependency signal is dominated by pan-essentiality rather than pair-specific interaction. The observation closest to our hypothesis is that tumor transcriptomic architecture reflects SL interactions [Haider et al., 2025]; we make the predictive direction explicit and, crucially, generate the transcriptome rather than requiring it to have been measured.

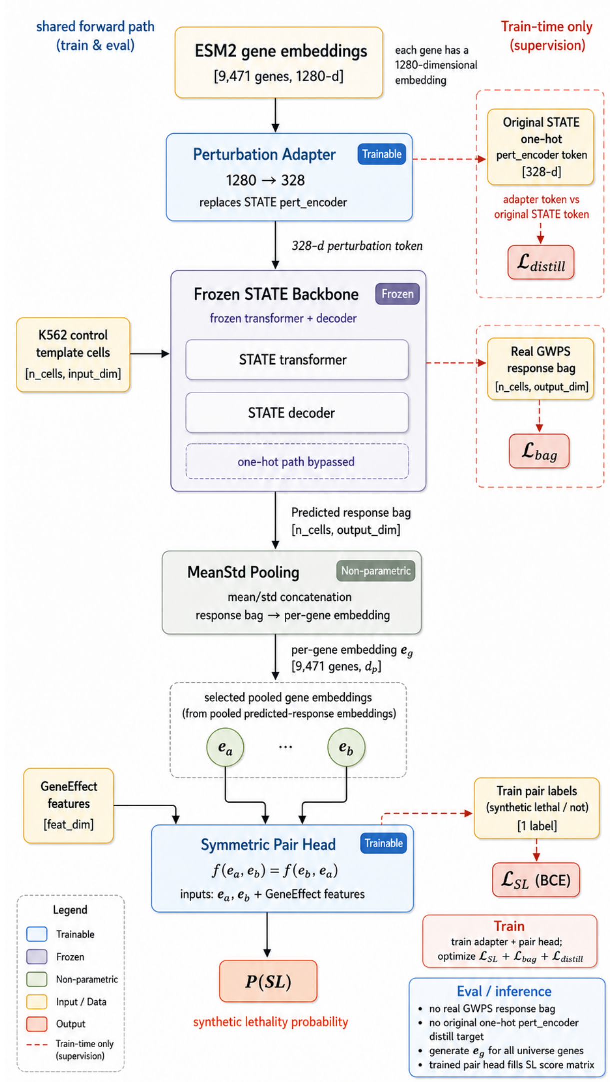

Figure 2: The end-to-end framework. A gene's ESM2 identity embedding is mapped by a trainable perturbation adapter into the perturbation space of a frozen STATE backbone, which generates a predicted response bag on a control-cell template. Mean–std pooling yields a gene embedding; a swap-invariant pair head produces the SL probability. Only the adapter and pair head are trainable (blue); the backbone is frozen (snowflake). Training combines an SL cross-entropy loss with a token-distillation loss and real-bag supervision.

3 Method

We cast synthetic-lethal (SL) candidate prioritization as ranking gene pairs by a predicted SL probability, where the features of each gene are derived from its *generated* perturbation response rather than from a pre-existing screen. Our central design choice is to obtain that response from a frozen virtual-cell foundation model driven by a protein-language-model gene identity, so that the pipeline can in principle score a gene that was never perturbed in the laboratory. Figure 2 shows the forward path. Section 3.1 fixes the prediction target and ranking protocol; Section 3.2 introduces the perturbation adapter that breaks the foundation model's closed vocabulary; Section 3.3 describes how a frozen backbone turns an adapted identity into a predicted response bag; Section 3.4 defines the swap-invariant pair head; and Section 3.5 gives the three-part training objective and the evaluation protocol, including an honest statement of how model selection interacts with the benchmark.

3.1 Problem setup and prediction target

We work on the K562-filtered, SynLethDB-derived benchmark of Feng et al. [2024], whose gene universe contains $9{,}471$ genes that carry numeric K562 DepMap GeneEffect values. A training example is an unordered gene pair (a, b) with a binary benchmark label $D_{ab} \in \{0, 1\}$ from a balanced Rand 1:1 split. For each gene g we also have its DepMap dependency scalar $C_g = \mathrm{GeneEffect}(\mathrm{K562}, g)$, a population-level fitness readout. The model outputs a probability $P(\mathrm{SL} \mid a, b)$, and we evaluate it with the benchmark's per-anchor protocol: for each anchor gene we rank its candidate partners and average ranking quality (NDCG@k, MAP@k) across anchors, alongside threshold-free classification (AUROC, AUPR). We report three splits of increasing cold-start severity: CV1 (random pair-level holdout), CV2 (one gene of each test pair held out of training), and CV3 (both genes held out). Because D is a benchmark-adapter label rather than a validated K562 SL assay, and its negatives are unconfirmed non-SL pairs, we treat all outputs as candidate prioritization, not target validation.

3.2 Perturbation adapter: breaking the closed vocabulary

The obstacle to scoring unscreened genes is that our virtual-cell backbone, Arc Institute's STATE model [Adduri et al., 2025], identifies a perturbation through a one-hot lookup over the fixed set of genes it was trained on. In our checkpoint that vocabulary covers only $1{,}542$ of the $9{,}471$ benchmark genes (16.3%), so a one-hot interface alone can never produce a response for the remaining 83.7% of the universe. We therefore replace the one-hot perturbation input with a learned adapter that maps a continuous, open-vocabulary gene identity into the backbone's perturbation space.

We represent every gene by its ESM2 protein-language-model embedding [Lin et al., 2023], a 1280-dimensional vector available for any protein-coding gene regardless of whether it was ever screened. The *perturbation adapter* is a two-layer multilayer perceptron, $\mathrm{Linear}(1280 \to 512) \to \mathrm{GELU} \to \mathrm{Linear}(512 \to d_p)$, whose output width d_p matches STATE's native perturbation-token dimension. Given a gene's ESM2 vector, the adapter emits a perturbation vector that is then consumed by STATE's own (frozen) perturbation encoder, exactly where the one-hot token would have entered. The advantage is reach: because the adapter's input is a smooth function of protein identity rather than a discrete index, the same trained module yields a perturbation token for genes the backbone never saw, which is the prerequisite for cold-start (CV3) scoring.

3.3 Generating the perturbation response with a frozen backbone

Given the adapted perturbation token, we generate the gene's response on a fixed template of 256 K562 control cells: STATE maps (control cells, perturbation token) to a predicted post-perturbation expression bag of the same shape, our in-silico analogue of a Perturb-seq readout. The entire STATE backbone is frozen (all parameters held at `requires_grad = False` and kept in evaluation mode); only the adapter and the downstream pair head are trained. Gradients still flow into the adapter through the frozen perturbation encoder, so the adapter learns to drive the backbone toward useful responses without disturbing the pretrained dynamics. Freezing is a deliberate data-efficiency choice: the benchmark provides only tens of thousands of labeled pairs, far too few to fine-tune a foundation model, whereas a small adapter and head are learnable from this signal while inheriting the backbone's pretrained perturbation biology.

3.4 Symmetric pair head

Synthetic lethality is an unordered relation, so the scorer must satisfy $f(a, b) = f(b, a)$; a head that treats (a, b) and (b, a) differently would learn spurious order artifacts. We first pool each predicted response bag into a gene embedding e_g by concatenating the per-feature mean and standard deviation across the 256 cells, which summarizes both the average shift and the dispersion of the generated response. We then combine the two gene embeddings through three element-wise swap-invariant blocks — the sum $e_a + e_b$, the absolute difference $|e_a - e_b|$, and the product $e_a \odot e_b$ — and append a small block of swap-invariant DepMap GeneEffect features (symmetric functions of C_a and C_b) together with a coverage indicator. A multilayer perceptron maps this representation to the pair logit, whose sigmoid is $P(\mathrm{SL} \mid a, b)$. Every operation upstream of the logit is symmetric in a and b, so the head is exactly swap-invariant by construction rather than by augmentation.

3.5 Training objective and evaluation protocol

The adapter and head are trained with a three-part objective that supervises the pipeline at the levels where signal is available. The primary term is a binary cross-entropy loss on the pair logit against the benchmark label D_{ab}. A *distillation* term keeps the adapter compatible with the pretrained backbone: for the in-vocabulary genes it minimizes the squared error between the adapter-produced perturbation token and STATE's own one-hot token, anchoring the open-vocabulary embedding to the space the backbone expects. A *bag supervision* term aligns the generated response with reality for genes that do have a real profile, minimizing the discrepancy (mean shift plus an energy distance) between the predicted and observed genome-wide Perturb-seq bags. To avoid leakage, bag supervision is applied only to training genes with a real profile; held-out genes are never shown a real bag. A short warmup emphasizes the distillation and bag terms before the SL term is activated, and the combiner drops any zero-weighted non-finite term so that an inactive loss cannot destabilize the gradient.

We deliberately state one limitation of the current evaluation. Because the benchmark protocol selects the best training epoch using the test-fold metric, our reported numbers are selection-matched to that protocol and are therefore not a strict embedding-only ablation against the dependency-only baseline. We report this openly and, in the experiments, treat the resulting numbers as preliminary single-fold status rather than as a completed cross-validated comparison.

4 Experiments

We organize the evidence around the paper's logic: first that a perturbation transcriptome predicts a gene's own dependency at all (the foundation), then that the observed transcriptome encodes a gene's SL partners when one partner is anchored (the proof-of-concept that licenses the method), then the current status of the generative framework, and finally a decomposition that explains why the cold-start residual is hard.

4.1 Setup

All SL experiments use the K562-filtered, SynLethDB-derived benchmark of Feng et al. [2024] (a $9{,}471$-gene universe, balanced Rand 1:1 pairs) under its per-anchor `cal_metrics` ranking protocol, reporting AUROC, AUPR, and NDCG@k/MAP@k. We use three splits of increasing cold-start severity: CV1 (random pair-level holdout), CV2 (one gene of each test pair held out), and CV3 (both genes held out). CV1 is reported only as a diagnostic: because positives and negatives differ in node degree, a probe using gene degree alone tops it (Table 2), so CV1 cannot demonstrate generalization to unseen genes. Generalization claims therefore come only from CV2 and CV3. The label D is a benchmark-adapter target whose negatives are unconfirmed non-SL pairs; we never interpret a high-ranked pair as a validated K562 SL target.

The observed-response experiments also face a concrete coverage bottleneck. The GWPS source used for transcriptome features provides real K562 CRISPRi response bags for $6{,}070/9{,}471$ benchmark genes (64.09%), while all other readable local CRISPRi/KO supplements increase the union to only $6{,}074$ genes (64.13%). At the pair level, the completed Rand 1:1 benchmark rows contain two GWPS-covered genes in only 51.17% of cases; the remaining rows require a fallback embedding for at least one gene. We therefore interpret transcriptome features as a test of the measured-data regime, not as evidence that observed Perturb-seq alone can solve the full cold-start task.

4.2 Foundation: the perturbation transcriptome predicts a gene's own dependency

Before asking whether a perturbation response names a gene's partners, we establish that it predicts the gene's *own* dependency (Table 1). Pseudobulk Δ-expression regressed onto DepMap GeneEffect reaches a 5-fold cross-validated Spearman of 0.485 on Replogle K562, and a same-cell-line transfer to the Adamson screen reaches AUROC 0.886 for strong dependencies; we treat the Adamson transfer as a same-line gate, not broad validation. An audit rules out the trivial explanation that the model only reads a generic cell-death axis: the death signature alone reaches Spearman 0.244, far below the transcriptome's 0.494, and residualizing the death signature out of the transcriptome leaves the signal essentially intact at 0.503. Single-cell distribution embeddings carry the same signal across datasets (exp03), though we flag that the strongest Adamson number there came from an Adamson-guided sweep and is exploratory, not a held-out result. Together these establish that the perturbation response encodes real, dependency-relevant transcriptional structure.

Table 1: Foundation: a gene's perturbation transcriptome predicts its *own* DepMap dependency, and the signal is real transcriptomic structure rather than a generic death axis. exp01 pseudobulk Δ-expression $\rightarrow$ GeneEffect; exp02 audit against the death signature; exp03 single-cell distribution embeddings (exploratory). The exp03 Adamson number came from an Adamson-guided sweep and is not a held-out generalization result.

Setting	Metric	Value
exp01 Replogle K562 (5-fold CV)	Spearman↑	0.485
exp01 Adamson same-line transfer	Spearman↑	0.500
exp01 Adamson same-line transfer	AUROC↑ (GE< -1)	0.886
exp02 death signature only	Spearman↑	0.244
exp02 best pseudobulk transcriptome	Spearman↑	0.494
exp02 death-residualized transcriptome	Spearman↑	0.503
exp03 scVI128 GMM-Ridge (Adamson sweep†)	Spearman↑	0.666
exp03 scVI128 GMM-Ridge (Adamson sweep†)	AUROC↑	0.911

†Adamson-guided sweep; exploratory, not held-out.

4.3 A dependency-only floor, and a degree-gameable diagnostic

We next fix the bar that any transcriptome-aware model must clear: a dependency-only classifier built from swap-invariant DepMap GeneEffect features (Table 2). On the honest splits this floor is already strong — CV2 AUROC 0.704 — which sets a demanding reference. The table also makes the CV1 caveat concrete: a degree-only probe wins CV1 ranking (NDCG@10 0.197) yet collapses to 0.001 on CV2 and CV3, confirming that CV1 rewards topology rather than biology. We therefore read CV1 as a sanity diagnostic and judge every subsequent result on CV2 and CV3.

Table 2: Dependency-only floor on the K562 SL benchmark (exp06), per-anchor `cal_metrics` ranking. CV1 (pair-level) is degree-gameable: a degree-only probe (shaded) wins it, so CV1 ranking is reported as a diagnostic, not as generalization. CV2 (one gene held out) and CV3 (both genes cold) are the honest surfaces.

	CV1 (diagnostic)		CV2 (one held out)		CV3 (both cold)	
Model	AUPRC↑	NDCG@10↑	AUPRC↑	NDCG@10↑	AUPRC↑	NDCG@10↑
Logistic reg.	0.648	0.004	0.648	0.005	0.645	0.004
XGBoost	0.812	0.050	0.732	0.042	0.609	0.002
Degree probe	0.902	0.197	0.750	0.001	0.750	0.001

4.4 Comparison with published SL-prediction methods

We place our label-free functional-signal methods next to four published synthetic-lethality predictors reproduced under the identical K562 protocol (same Rand 1:1 splits, same per-anchor `cal_metrics`; Table 3). DDGCN and SLGNN are graph-neural-network link predictors; GRSMF and SL2MF are matrix-factorization models. All four *consume the SL association matrix as input*; our methods use only GeneEffect and perturbation-response features and see no SL labels at feature-construction time.

Two patterns are honest to report. First, on ranking the label-graph methods lead: GRSMF reaches a cross-CV NDCG@10 of 0.317 and DDGCN 0.257, against 0.032 for our dependency-only floor and 0.090 once the observed transcriptome is added. The gap persists at cold-start CV3, so we make no claim of a cold-start advantage. Second, on classification the picture is closer: our dependency-only floor sits at a cross-CV AUROC near 0.70, within range of DDGCN and GRSMF (0.752) and SLGNN (0.721). The best classifier (SLGNN, mean F1 0.741) is not the best ranker (GRSMF), which is why we report both F1 and NDCG@10 throughout. These published methods set the ranking bar that our generative direction (Section 4.6) is intended to close, without yet doing so.

Table 3: Synthetic-lethality prediction on the K562 SL benchmark: published label-graph methods (which consume the SL association matrix as input; Cai et al., 2020, Huang et al., 2019b, Liu et al., 2020, Zhu et al., 2023) versus our label-free functional-signal methods. Same K562-DepMap Rand 1:1 splits and identical per-anchor `cal_metrics` ranking on both sides. Bold marks the best value in each metric column, including ties. Label-graph methods lead on ranking; our functional methods use no SL labels yet stay classification-competitive, leaving ranking as the open gap. CV1 is the degree-gameable diagnostic.

	CV1 (diag.)		CV2		CV3		Cross-CV mean			
Method	F1↑	N@10↑	F1↑	N@10↑	F1↑	N@10↑	AUROC↑	AUPR↑	F1↑	N@10↑
Label-graph methods (consume the SL matrix as input)										
DDGCN	0.822	0.286	0.685	0.241	0.606	0.243	**0.752**	0.743	0.704	0.257
GRSMF	0.774	**0.322**	0.658	**0.315**	0.625	**0.313**	**0.752**	0.771	0.686	**0.317**
SL2MF	0.666	0.278	0.422	0.127	0.288	0.074	0.456	0.448	0.459	0.160
SLGNN	**0.870**	0.150	0.685	0.050	0.667	0.000	0.721	0.746	**0.741**	0.066
Functional-signal methods (label-free)										
Dependency-only (exp06)	0.730	0.050	0.676	0.042	0.670	0.002	0.698	0.718	0.692	0.032
+ transcriptome (exp07)	0.769	0.174	**0.702**	0.094	**0.676**	0.001	0.744	**0.772**	0.716	0.090

4.5 Proof-of-concept: the observed transcriptome encodes SL partners

The pivotal question is whether the observed perturbation transcriptome carries partner information beyond what dependency scores already provide. Holding the harness, seeds, and splits fixed and changing only the feature matrix, adding the observed transcriptome more than doubles CV2 ranking quality (NDCG@10 $0.042 \rightarrow 0.094$; AUROC $0.704 \rightarrow 0.751$; Table 4), and the lift concentrates on rows where a real profile can anchor the pair. This is the empirical license for the whole approach: when one partner is measured, its response encodes information about its synthetic-lethal partners. The same table delivers the matching bad news. Coverage-limited observed features do not rescue CV3, where both genes are cold to the SL training graph and the top-k ranking lift vanishes (Figure 1 and the difficulty ladder, Figure 3). Cold-start ranking is therefore the open frontier, and it is exactly the regime an observed-only method cannot reach reliably, because many candidate genes have no real shockwave to read.

Table 4: Adding the observed perturbation transcriptome to the dependency-only features (exp07; identical harness, seeds, and splits, only the feature matrix changes). On CV2, where one partner is anchored by a real Perturb-seq profile, transcriptome features more than double ranking quality (NDCG@10 row, shaded). On CV3 (both genes cold, no observed profile) the lift vanishes. `full_universe` = all benchmark pairs; `covered_pairs` = pairs whose anchored gene has a real profile.

Split	Features	AUROC↑	NDCG@10↑	MAP@10↑
CV2	dependency-only	0.704	0.042	0.034
CV2	+ transcriptome	0.751	0.094	0.079
CV3	dependency-only	0.596	0.002	0.001
CV3	+ transcriptome	0.630	0.001	0.001

4.6 The generative framework: preliminary status

We now report the current status of the framework that targets the CV3 frontier by *generating* responses for unscreened genes (Table 5). These results are preliminary. The runs are incomplete (single selected folds, not a 5-fold mean), the best-epoch selection reads the test fold (Section 3.5), and the best folds do not yet clear the dependency-only floor (CV2 AUROC 0.667 versus the floor's 0.704). We make no claim that the framework beats any baseline; we report it as the present state of a newly proposed method whose tuning is ongoing. What the experiment does establish is that the pipeline runs end to end and produces calibrated SL probabilities for genes outside STATE's 16.3% in-vocabulary set, which an observed-transcriptome method structurally cannot do.

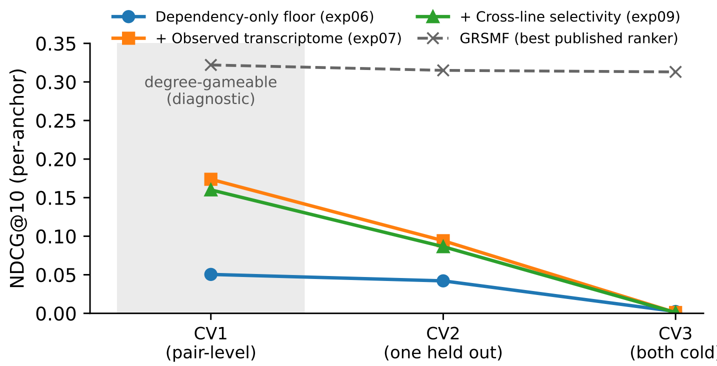

Figure 3: Ranking quality down the cold-start difficulty ladder. The observed transcriptome lifts CV2 (one gene anchored) but every signal — dependency, transcriptome, and cross-line selectivity — collapses on CV3 (both genes cold). CV1 is shaded as the degree-gameable diagnostic.

Table 5: **Preliminary** results for the virtual-cell framework (exp08, AIVC/STATE+ESM2 adapter). Values are the *best selected fold*, *not* a 5-fold mean, and best-epoch selection reads the test fold; full 5-fold tuning is ongoing (see Limitations). The dependency-only floor (exp06) is shown only for reference. We make no claim that the framework beats the floor; these numbers report the current status of a newly proposed method.

Split	Method	AUROC↑	NDCG@10↑
CV2	dependency-only floor (ref.)	0.704	0.042
CV2	AIVC/STATE+ESM2 (best fold, prelim.)	0.667	0.005
CV3	dependency-only floor (ref.)	0.596	0.002
CV3	AIVC/STATE+ESM2 (best fold, prelim.)	0.587	0.003

4.7 Decomposition: most cold-start signal is pan-essentiality

Finally we explain why the cold-start residual is so hard (Table 6). Cross-cell-line selectivity features lift classification on both honest splits (CV2 AUROC $0.704 \to 0.742$, CV3 $0.596 \to 0.645$), which might suggest progress on co-dependency. But when the benchmark is restricted to non-pan-essential pairs on CV3, performance falls back toward chance (AUROC 0.583, AUPR 0.490). The lift was therefore carried mostly by pan-essentiality structure — genes that are broadly essential are easy to flag — while pair-specific synthetic lethality, the biologically interesting relation, remains the thin and unsolved residual. This reframes the cold-start gap as a precise target: a generative virtual-cell model must learn the pair-specific signal that genome-wide essentiality gravity hides, which is exactly what the framework in Section 3 is designed to pursue.

Table 6: Cross-cell-line selectivity features (exp09). They lift classification on both honest splits, but when the benchmark is restricted to non-pan-essential pairs on CV3 (shaded), AUROC and AUPR collapse toward chance. Most of the genome-wide co-dependency signal is pan-essentiality structure; pair-specific synthetic lethality is the thin, hard residual.

Split	Features	AUROC↑	AUPR↑	NDCG@10↑
CV2	dependency-only	0.704	0.732	0.042
CV2	+ selectivity	0.742	0.769	0.086
CV3	dependency-only	0.596	0.609	0.002
CV3	+ selectivity	0.645	0.651	0.001
CV3	+ selectivity, non-pan-ess.	0.583	0.490	0.001

5 Conclusion

We studied whether a gene's perturbation transcriptome encodes its synthetic-lethal partners, and how to extend that signal to genes that were never screened. On a SynLethDB-derived K562 benchmark with a leakage-controlled difficulty ladder, we showed that the observed perturbation transcriptome more than doubles partner-ranking quality when one partner is anchored by a real Perturb-seq profile, establishing that the same shockwave which reads a gene's own dependency also carries information about its partners. To reach genes with no observed response, we proposed an end-to-end framework that generates the response from a frozen virtual-cell foundation model conditioned on a protein-language-model gene identity, breaking the backbone's closed vocabulary. A decomposition then clarified why the cold-start residual is hard: most genome-wide co-dependency signal is pan-essentiality, leaving pair-specific synthetic lethality as a thin residual that a generative model must isolate.

The main limitation is not only model maturity but data coverage. Replogle K562 GWPS substantially expands the measured response library, yet it still covers only $6{,}070/9{,}471$ benchmark genes, and roughly half of the evaluated pair rows lack real GWPS profiles for both genes. This makes the observed-transcriptome result best understood as an anchored-regime finding: it shows that a measured shockwave carries partner signal when coverage exists, but it also explains why observed Perturb-seq alone cannot solve both-gene-cold ranking. The generative framework is therefore motivated by a real data bottleneck, although its current runs are preliminary, incomplete, and do not yet clear the dependency-only floor, so we make no superiority claim.

Several broader limitations also bound these claims. All results are for a single cellular context (K562) and a benchmark-adapter label whose negatives are unconfirmed non-SL pairs, not a validated K562 SL assay, so high-ranked pairs are candidates for follow-up rather than confirmed targets. Cold-start ranking (both genes unseen) remains unsolved by every signal we tested. The natural next steps are to complete cross-validated tuning of the generative framework with an evaluation that does not select on the test fold, to extend beyond K562 to cross-lineage and patient-context transfer, and to supervise the model directly on the pair-specific residual that pan-essentiality otherwise hides.

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A Artifact and reproducibility notes

All quantitative values in the tables are extracted directly from on-disk experiment artifacts by a single tested script (`docs/report/scripts/make_tables.py`); no number is entered by hand. The benchmark splits, per-anchor ranking protocol, and feature definitions follow the experiment logs referenced in each table caption.